**25900686**

**MAHATMA GANDHI UNIVERSITY, KOTTAYAM**

**MGU-UGP (HONOURS) Regular EXAMINATION October 2025**

**Third SEMESTER**

**Discipline Specific Elective (DSE) - MG3DSEZGY200 - ETHOLOGY**

**(2024 ADMISSION ONWARDS)**

**Duration: 2 Hours** **Maximum Marks: 70**

**Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)**

**Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.**

## Part A

**Multiple Choice Questions Answer all questions; Each question carries 1 marks**

**1. Cerebrum is a part of……. [K] / [CO2]**

**a). Forebrain**

**2. Almond shaped structure in brain is…… [K] / [CO2]**

**b). Amygdala**

**3. Which reproduction strategy would be favored in a resource rich environment?**

**b). k-selection strategy [An] / [CO3]**

**4. Trophallaxis in ants is: [K] / [CO4]**

**b). Exchange of food mouth-to mouth**

**5. Which hormone is known as the 'stress hormone'? [U] / [CO3]**

**b). Cortisol**

**6. Which motor area is responsible for programming sequences of movement? [K] / [CO2]**

**a). Supplementary motor area**

**7. Select the primary cost associated with large groups. [K] / [CO4]**

**b). Higher competition for food**

**8. Male-male competition is usually seen in which type of sexual selection? [U] / [CO3]**

**c). Intrasexual selection**

**9. Illustrate the adaptive value of territorial marking. [K] / [CO4]**

**c). Reduces direct conflict and saves energy**

**10. Ethology helps in animal reintroduction programs by: [K] / [CO1]**

**c). Both A and B [1x10 = 10]**

## Part B Short Answer Question

**Answer any 10 questions Each question carries 3 marks**

**11. Compare the role of Estrogen and testosterone in parental behaviour. [An] / [CO3]**

Estrogen**,** primarily a female hormone, enhances nurturing and caregiving instincts by stimulating maternal behaviors such as nursing, grooming, and protection of offspring. It acts on brain regions like the hypothalamus and amygdala to increase sensitivity to infant cues and promote bonding. In contrast**,** testosterone**,** predominantly a male hormone, is often associated with aggression, territoriality, and mating effort, but in some species, moderate levels of testosterone can also support paternal care by motivating protection and defense of offspring. After birth, testosterone levels typically decrease in males involved in caregiving, allowing more nurturing behaviors to emerge. Thus, estrogen generally promotes caregiving and bonding**,** while testosterone influences protection and parental investment**,** balancing reproductive and parental roles between the sexes.

***(Give full mark for relevant points; 1.5 marks for Each Hormone; Liberal valuation)***

**12. Differentiate between explicit and implicit memory [U] / [CO2]**

Explicit memory is the conscious, intentional recall of facts and personal experiences, while implicit memory is the unconscious, automatic recall of skills, habits, and conditioned responses. Explicit memory requires conscious effort, like remembering a name or an event, whereas implicit memory influences behavior without conscious thought, such as riding a bike.

***(Give full mark for relevant points; 1.5 marks for Each)***

**13. Explain the two major mechanisms behind coloration in animals [U] / [CO3]**

Animals exhibit coloration through two major mechanisms — pigmentary coloration and structural coloration**.** Pigmentary coloration results from chemical pigments such as melanin, carotenoids, and pteridines that absorb certain wavelengths of light and reflect others, producing colors like browns, reds, and yellows. Structural coloration, on the other hand, arises from microscopic structures on the surface of the skin, scales, or feathers that interfere with light, producing iridescent or metallic colors such as the blue of a peacock's feather or a butterfly's wing. Together, these mechanisms play vital roles in camouflage, communication, thermoregulation, and mate attraction.

(***Liberal valuation; mechanism not in syllabus; if different types included give full credits)***

**14. Give any two examples where reciprocal altruism is observed. [U] / [CO4]**

Reciprocal altruism is a behavior in which one animal helps another with the expectation that the favor will be returned in the future. It is based on mutual benefit over time.

In vampire bat colonies, not all individuals successfully find food every night. A well-fed bat often regurgitates a small amount of blood to feed a hungry companion that failed to feed. Later, when the roles reverse, the previously helped bat will share food in return. This behavior increases survival chances for all members and strengthens social bonds.

Primates often groom one another, removing parasites and dirt. Grooming not only provides hygiene benefits but also strengthens alliances. An individual who grooms another today is likely to receive grooming, protection, or food sharing in return later.

In lions, males within a coalition cooperate to defend a pride's territory. Even if one male benefits immediately (e.g., by mating), others help because they expect similar support in the future. This mutual cooperation enhances the survival and reproductive success of the group.

***(Give full mark for relevant points and any two examples; 1.5 mark for Example)***

**15. Discuss learning centres in brain. [U] / [CO2]**

Learning involves several interconnected regions of the brain. The hippocampus plays a major role in forming new memories and spatial learning. The amygdalais involved in emotional learning, especially fear and reward associations. The cerebral cortex**,** particularly the prefrontal cortex**,** supports higher learning functions such as decision-making and problem-solving. Together, these centers coordinate the storage, retrieval, and modification of learned information.

***(Mention any 3 centres, give full credits)***

**16. How did I.P.Pavlov's work show the role of conditioning in shaping behavior? [U] / [CO1]**

I.P. Pavlov demonstrated classical conditioningthrough his experiments with dogs. He showed that dogs could learn to associate a neutral stimulus, like a bell, with food, and eventually salivate to the sound alone. This proved that behavior can be modified through learned associations between stimuli, establishing the foundation of behavioral psychology and showing that learning is not purely instinctive but can be conditioned.

***(Give full mark for relevant points)***

**17. What motor deficits occur with damage to the premotor cortex? [K] / [CO2]**

Damage to the premotor cortex can cause difficulties in planning and coordinating voluntary movements. A person may retain strength but struggle to perform complex or sequential actions (a condition known as apraxia). It also impairs the ability to use sensory cues to guide movements, such as reaching accurately for an object or imitating gestures.

***(Liberal Valuation; Give full mark for relevant points)***

**18. How does group foraging differ from solitary foraging? [U] / [CO4]**

In group foraging, animals search for food together, which can improve hunting efficiency, predator detection, and access to information about food sources. However, it also leads to competition for resources. In solitary foraging**,** individuals search alone, reducing

competition and allowing independence, but increasing vulnerability to predators and reducing foraging success for large or difficult prey.

***(Give full mark for relevant points with examples; 1.5 mark for Each)***

**19. A male bird continues to sing despite the increased risk of attracting predators. Analyze this behavior in terms of the trade-off between survival and reproductive success. What does this suggest about the evolutionary pressures shaping communication strategies in animals?**
**[An] / [CO5]**

A male bird singing despite the risk of attracting predators demonstrates an evolutionary trade-off between survival and reproductive success. While singing increases predation risk, it also enhances the male's chances of attracting mates and defending territory. Natural selection favors individuals that balance this risk effectively — those whose song behavior results in greater reproductive success, even at some survival cost. This suggests that evolutionary pressures shape communication strategies to maximize reproductive fitness, even when they involve survival risks. ***(Give full mark for relevant points)***

**20. Compare aestivation and hibernation. [An] / [CO3]**

Aestivationand hibernation are both states of dormancy, but they occur in different seasons and serve different purposes. Aestivation occurs during hot and dry periods to prevent desiccation and conserve water (e.g., in snails, lungfish, and frogs). Hibernation, on the other hand, occurs during cold seasons to conserve energy when food is scarce (e.g., in bears and bats). Both reduce metabolic rate, heart rate, and body activity to enhance survival under extreme environmental conditions.

***(Give full mark for relevant points with examples; 1.5 mark for Each)***

**21. Interpret EEG in REM vs NREM sleep. [U] / [CO3]**

During NREM (Non-Rapid Eye Movement) sleep, the EEG shows slow, high-amplitude delta waves**,** indicating deep sleep and reduced brain activity. In contrast, REM (Rapid Eye Movement) sleep displayslow-amplitude, high-frequency waves similar to the awake state, reflecting intense brain activity and vivid dreaming. Thus, EEG patterns reveal that the brain is highly active during REM but restful during NREM. ***(Give full mark for relevant points; 1.5 mark for Each).***

**22. Critically analyze limitation of Lorenz's psycho-hydraulic model. [U] / [CO2]**

Lorenz's psycho-hydraulic model explains motivation as a build-up of internal "energy" that must be released through behavior, like water overflowing from a tank. However, it oversimplifies behavior by ignoring cognitive control, learning, and environmental influences**.** It assumes fixed patterns of motivation and doesn't account for flexibility or choice in behavior. Modern ethology recognizes that motivation is influenced by both internal states and external stimuli**,** making Lorenz's model useful historically but limited scientifically. ***(Give full mark for relevant points).***

**Part C Short Essay Questions Answer any 6 questions Each question carries 5 marks**

**23. Analyze how male scorpions use behavioral adaptations to safely engage with potentially aggressive females. [A] / [CO3]**

Male scorpions use several behavioral adaptations to safely approach and mate with potentially aggressive females. During courtship, the male performs a distinctive "dance", in which he grasps the female's pincers and guides her in a coordinated movement. This behavior helps the male assess the female's receptivity and reduce her aggression. He also deposits a spermatophore on the ground and carefully maneuvers the female over it to achieve fertilization without direct copulation, minimizing risk of injury or cannibalism. These adaptations reflect the balance between reproductive success and self-preservation in scorpion mating behavior. ***(Detailed explanation give full mark).***

**24. Explain different grouping patterns in primates [K] / [CO4]**

Southwick abd Siddique ((1974) outlined social behaviour in primates into six types

- **Type 1 Solitary:** Individuals of a species forage and live alone, with males often having territories that encompass several females. Example: Orangutans
- **Type 2 (Monogamous):** A single adult male and female live together with their offspring. They are often highly territorial and may mate for life.Eg: Gibbon species
- **Type 3 One male groups with bonded female and offsprings:** A single adult male lives with multiple adult females and their offspring. This is common in species where males compete to gain control of a group. **Example:** Hanuman Langur.
- **Type 4 Aggregate single male group with bonded female and offsprings:** A single adult male lives with multiple adult females and their offspring. This is common in species where males compete to gain control of a group.**Example:** Hamadrays Baboons.
- **Type 5 Multi-male bisexual groups:** Groups consist of multiple adult males and females, along with their offspring. Eg. Gorillas, Macaques. These groups often have a stable core of related females, and males typically leave their birth group upon reaching maturity.A clear dominance hierarchy, especially among males, determines access to resources and mates.
- **Type 6 Diffuse social parties:** A large community of individuals splits into smaller subgroups (fission) for foraging or other daily activities and then reunites (fusion) at a common sleeping site. Chimpanzees. ***(Explanation of any 5 types-; 1 mark each).***

**25. Explain latent learning in the light of Tolman's experiment. [U] / [CO2]**

Latent learning refers to learning that occurs without any immediate reinforcement or obvious change in behavior, becoming evident only when a motivation or reward is introduced. Edward C. Tolman demonstrated this concept through his famous maze experiment with rats. In his study, one group of rats was rewarded with food for reaching the end of a maze each day, while another group received no reward. Initially, only the rewarded rats showed improved performance. However, when the unrewarded rats were later given food at the end of the maze, they quickly navigated it as efficiently as the rewarded group. This showed that the rats had learned the layout (a cognitive map) of the maze earlier but had not demonstrated

it until motivated by a reward. Tolman's experiment proved that learning can occur without reinforcement, emphasizing the role of cognitive processes in behavior.

***(Latent learning 2 mark; Experiment 3 marks).***

**26. Compare and contrast Batesian and Müllerian mimicry, highlighting their differences and similarities. [An] / [CO3]**

Batesian and Müllerian mimicry are both evolutionary strategies where one species resembles another to gain protection from predators, but they differ in the nature of the mimicry. In Batesian mimicry, a harmless species imitates the warning signals of a harmful or unpalatable species to deceive predators; for example, the harmless viceroy butterfly mimics the toxic monarch butterfly. In Müllerian mimicry, two or more genuinely harmful or unpalatable species resemble each other, reinforcing the avoidance behavior in predators; for instance, several species of bees and wasps share similar black and yellow warning patterns. While both forms of mimicry involve visual resemblance and predator learning, Batesian mimicry is a one-sided advantage for the mimic, whereas Müllerian mimicry benefits all species involved by strengthening the predator's avoidance response.***(Explain both mimicry 2 marks;* differences and similarities 3 marks*).***

**27. Explain in detail on Taste aversion with examples. [K] / [CO2]**

Taste aversion is a type of associative learning where an animal learns to avoid a food that previously caused illness or discomfort. It is a form of classical conditioning, but it is unique because it can occur after just one pairing of a particular taste with nausea and can develop even with a long delay between eating and feeling sick. This adaptation helps animals avoid toxic or spoiled foods in the future, thus increasing survival chances.

In a famous experiment by Garcia and Koelling (1966), rats were given sweet-tasting water followed by exposure to radiation that caused nausea. Later, the rats avoided the sweet water, associating its taste with illness. In nature, wolves or coyotes may develop taste aversion to livestock meat if it is laced with a mild toxin during conservation training. Thus, taste aversion demonstrates how animals quickly learn to link taste cues with negative physiological outcomes.***(Explanation 3 mark; Examples 2 marks).***

**28. Compare different types of parenting in animals. [An] / [CO3]**

Parenting in animals varies widely depending on species, ecology, and reproductive strategy.

Maternal care: The female alone cares for the offspring, as seen in mammals, where mothers nurse and protect young.

Paternal care: The male provides most of the care, common in some fish and birds like the stickleback and emu.

Biparental care: Both parents share the responsibility, seen in many birds such as penguins and doves, ensuring high offspring survival.

Alloparental care: Individuals other than parents (like siblings or group members) help in raising the young, observed in lions and wolves.***(Definition -1 mark, each types 1 marks).***

**29. Discuss various biological aspects of pheromons. [U] / [CO5]**

Pheromones are chemical substances secreted by animals that influence the behavior or physiology of others of the same species. They are crucial in communication and play roles in reproduction, defense, and social organization.

Types and functions:Sex pheromones: Attract mates and signal readiness for reproduction (e.g., female moths release pheromones to attract males).

Alarm pheromones: Trigger escape or defense responses in group members (e.g., ants release alarm pheromones when attacked).

Trail pheromones: Help mark paths to food sources, as in ants and termites.

Territorial pheromones: Used to mark boundaries and prevent intruders (e.g., in dogs and tigers).

Social pheromones: Regulate colony structure in social insects like honeybees, where the queen's pheromones maintain social hierarchy.

Biologically, pheromones are detected by specialized receptors in the olfactory system or vomeronasal organ (VNO) and trigger innate behavioral or hormonal responses, ensuring coordination and survival within a species.***(Definition -1 mark, any 4 types 1 mark each).***

**30. Analyse tidal rhythms in coastal organisms and connect them to survival strategies.**

**[An] / [CO3]**

Tidal rhythms are behavioral and physiological cycles in coastal organisms that correspond to the rise and fall of tides. These rhythms are controlled by internal biological clocks synchronized with tidal cues such as water movement, pressure, and light changes. Many coastal animals, like crabs, mussels, and mollusks, time their activities—feeding, reproduction, and movement—to these tidal cycles. For example, fiddler crabs emerge from burrows during low tide to feed on exposed mudflats and retreat during high tide to avoid predators and strong currents. Similarly, mussels close their shells during low tide to prevent desiccation. These adaptations help organisms optimize resource use, avoid predation, and cope with environmental fluctuations. Thus, tidal rhythms play a vital role in enhancing survival and reproductive success in intertidal habitats***.((Definition -2.5 mark, explanation with relevant example 2.5 marks).***

**[5x6 = 30]**